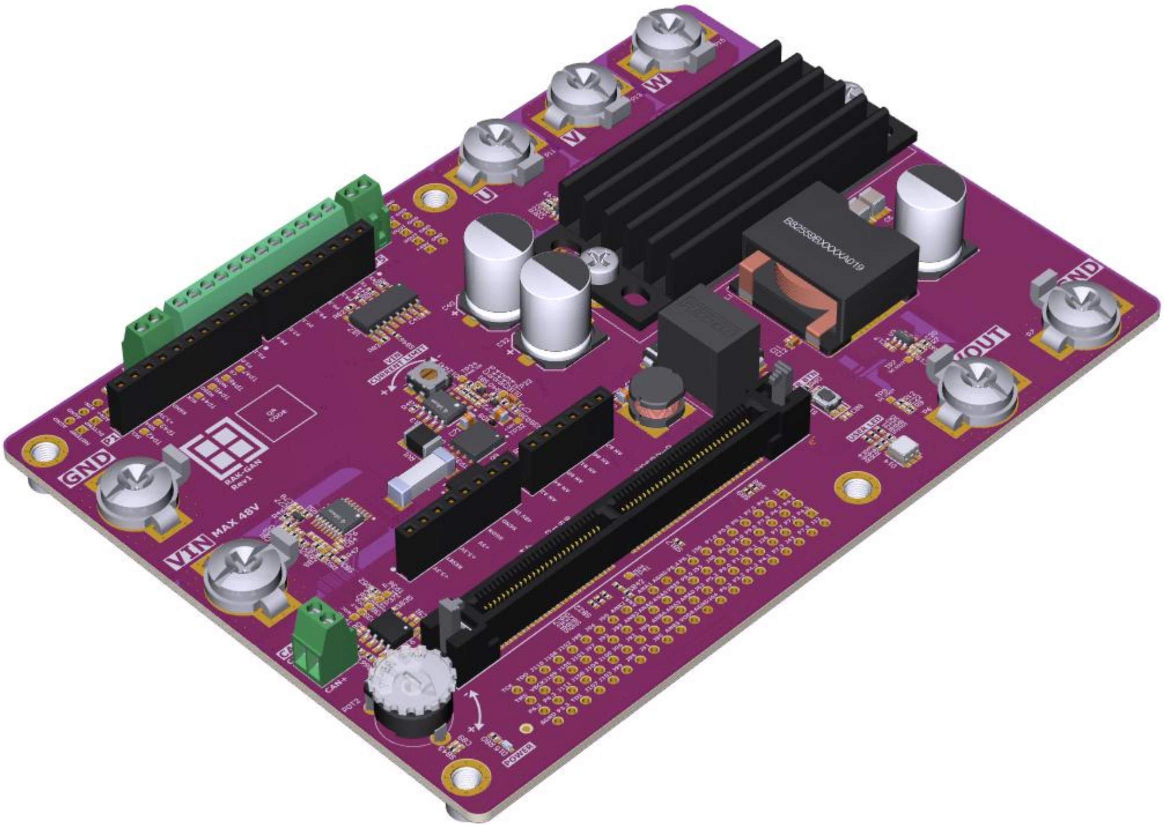
RAK-GaN

User Manual



Versions

Version	Date	Rationale
0.1	March 25, 2026	First draft. Author: GDR
1.0	June 1, 2026	Release. Author: GDR, KOA

Legal Disclaimer

Table of Contents

Overview

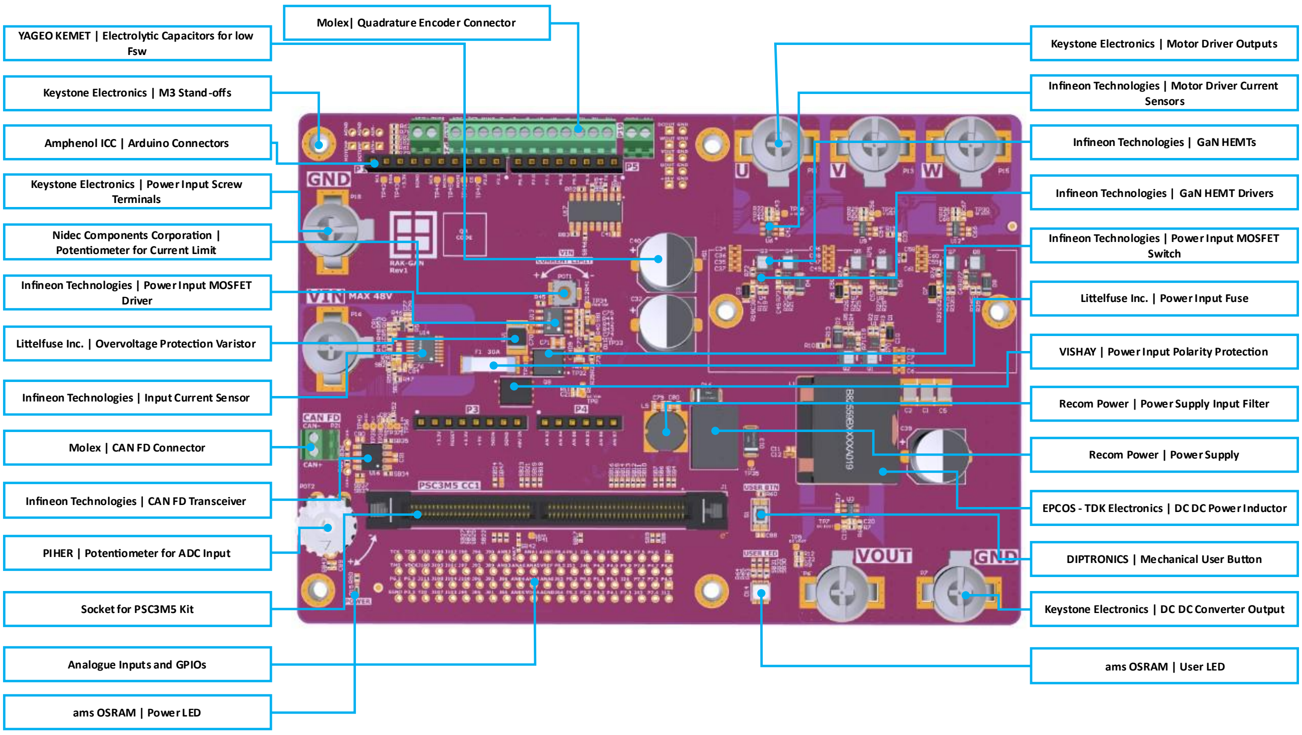
Features

The RAK-GaN [Rutronik Adapter Kit Gallium Nitride] is an independently developed evaluation platform which requires the Infineon KIT PSC3M5 CC1 board. The features of the KIT PSC3M5 CC1 are enhanced by the RAK-GaN evaluation platform, creating a flexible development platform for power conversion and motor control applications.

Key features of RAK-GaN:

- KIT_PSC3M5_CC1 - Infineon's Devkit with PSOC™ Control C3M5 MCU onboard.
- IGB070S10S1XTMA1 - Infineon's CoolGaN™ Transistors for DC/DC Converter and Motor Driver Bridges.
- ISC035N10NM5LF2 - Infineon's OptiMOS™ 5 Linear FET 2 Transistor for Power Input Control.
- 1EDN7136UXTSA1 - Infineon's High-side TDI GaN SG HEMT gate drivers.
- 1EDL8011XUMA1 - Infineon's EiceDRIVER™ 125 V High Side Gate Driver for Power Input Control transistor.
- TLE5571 and TLE5572 - Infineon's TMR Current Sensors.
- TLE9371VSJXTMA1 - Infineon's Signal improvement transceiver as CAN FD Driver.
- R-78HE5.0-0.3 - RECOM's 5V 300mA Power Supply Module.
- V30K202-M3/H - VISHAY Schottky Diode for power input polarity protection.
- B82559B5472A016 – EPCOS High Power Inductor.
- 39773-000**x** - Molex terminal blocks for CAN FD and Quadrature Encoder.
- SK DC 16 58 SA - Extruded heatsink from Fischer Elektronik.
- P/N: 1202 – 30A Screw Terminals from Keystone Electronics.
- 75915-3**xx**LF – Arduino Connectors from Amphenol ICC.
- 0463030.ER and V56MLA1812NRAUTO – Fuse and Varistor for power input protection from Littelfuse Inc.
- E3635 - ams OSRAM RGB User LED OSIRE®.
- PZS525V1 and PJE8408 – Zener Diodes and MOSFET from PANJIT Inc.
- A786MS107M1JLAS015 - Aluminum Polymer Capacitor Radial from Yageo KEMET.

Component Placement



Hardware

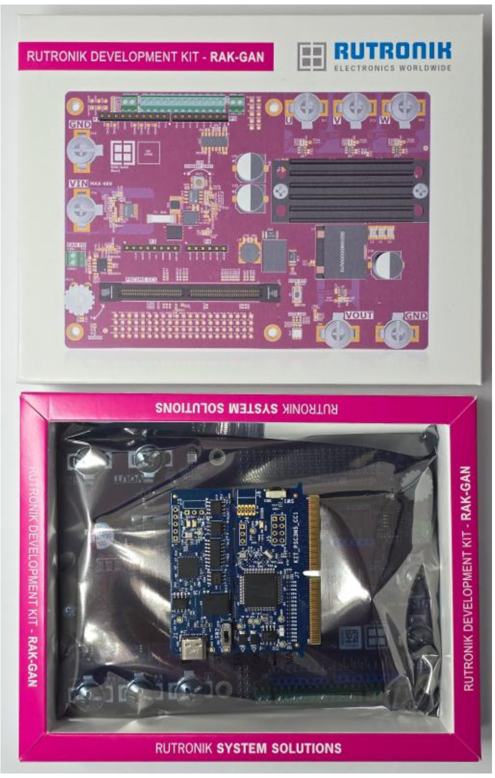
Electrical Specifications

Parameter	MIN	TYP.	MAX
VIN Power Input Voltage	10V		48V
VIN Power Input Current			30A[1]
VOUT DC/DC Converter Current			19A[2]
DC/DC Converter Switching Freq.		400kHz	
Motor Driver Output Voltage			48V
Motor Driver Output Current[3]			25A
Motor Driver Switching Freq.		100kHz	200kHz
Motor FOC Bandwidth			4kHz[4]

Delivery Set

The RAK-GaN package includes:

- The RAK-GaN board.
- The KIT_PSC3M5_CC1 board.



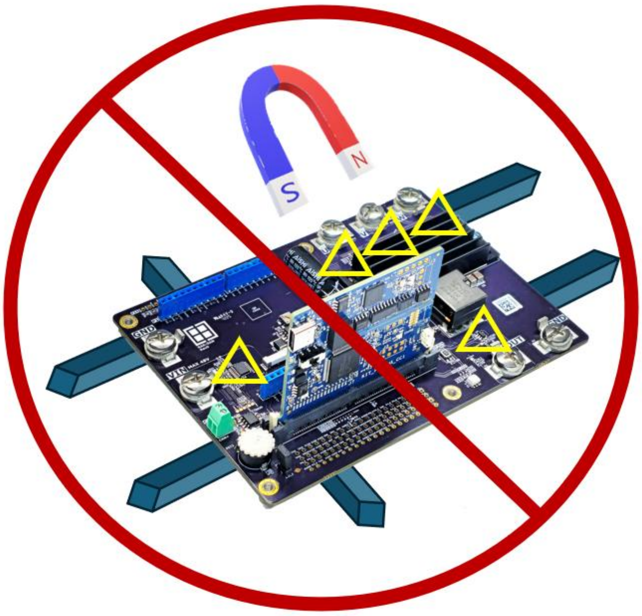
[1] Fuse F1 operating time – 4 Hours at 30A.
[2] Limited by L1 current capability.
[3] Active cooling required.
[4] FOC loop execution every PWM cycle.

Cable Requirements

The cables are not included in a box. If you are developing an application where currents are up to 30A, at least 3 mm^2 (12 AWG) cable is required.

TMR Current Sensors

The RAK-GaN board has 4 TLE5572 and 1 TLE5571 TMR sensors. These sensors are sensitive to strong magnetic fields that may be present near the board. Always check for massive magnetic objects nearby when developing with RAK-GaN.



Magnetised objects can disrupt the operation of TMR current sensors

Contactless current sensing design

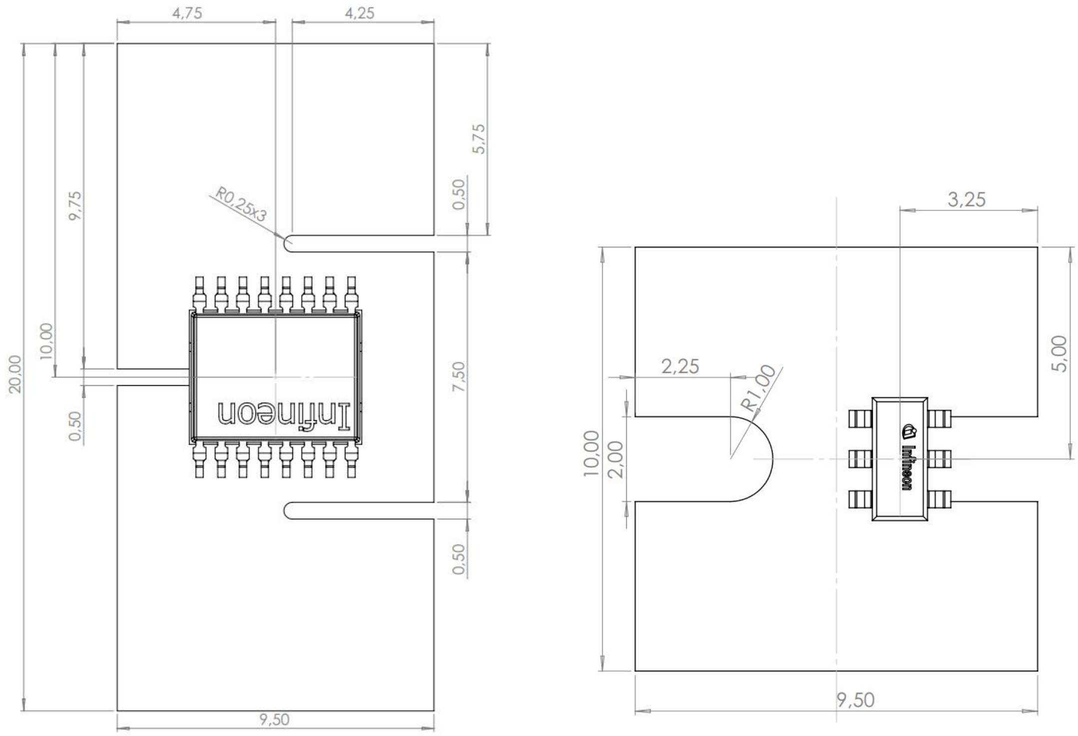
The TMR current sensors are used in the RAK-GAN project to measure the main power input current, the DC/DC converter output current, and the currents of each motor driver phase. The TLE5571 and TLE5572 both offer high sensitivity and accuracy, along with a wide dynamic range and bandwidth. The TLE5571 has more safety features that are also enabled in the RAK-GAN design.

U-Shape and Straight Power Rails

The TLE5571 is mounted on a U-shaped power rail; meanwhile, the TLE5572 is mounted on a straight power rail. Both rails are formed from stacked PCB traces, and both sensors are positioned into a „sweet spot“ where magnetic strays are the most intense and have the closest proximity to the sensing area of the chip. The TLE5571 has a bandwidth of 2

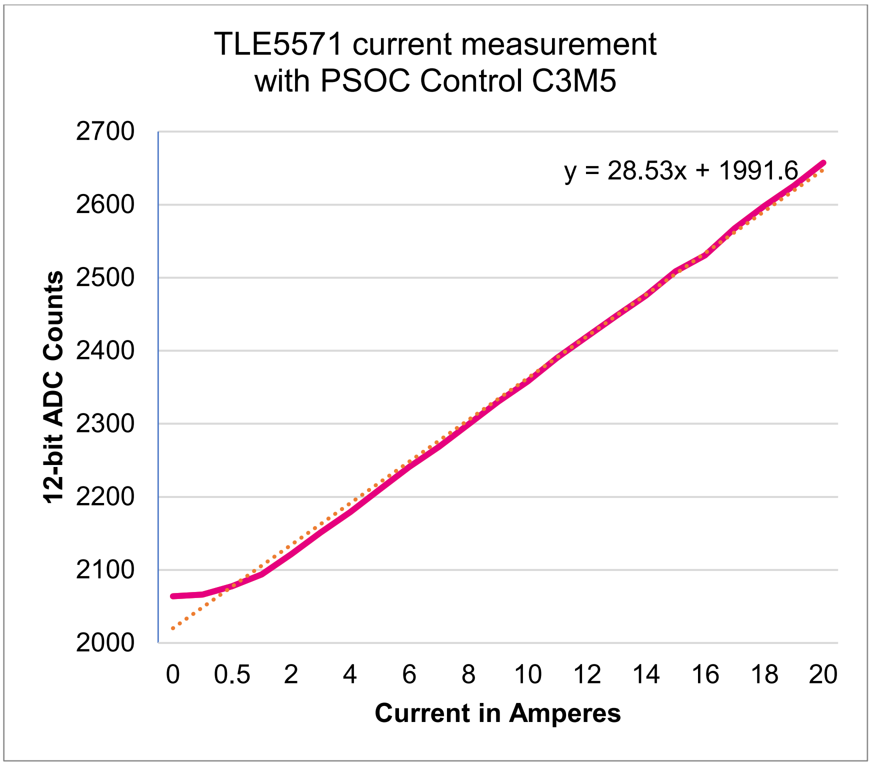
MHz, 4dB overshoot, and a sensitivity of 61 μT/A. The TLE5572 has been positioned for the best bandwidth, which is 850kHz at 69.9 μT/A, see the table below.

Position	Bandwidth [kHz] (-3dB)	μT/A
middle	250	89.3
+0.5mm from the middle	370	87.4
+0.7	400	85.5
+0.9	500	82.8
+1.1	600	79.3
+1.3	750	75.0
+1.5	850	69.9

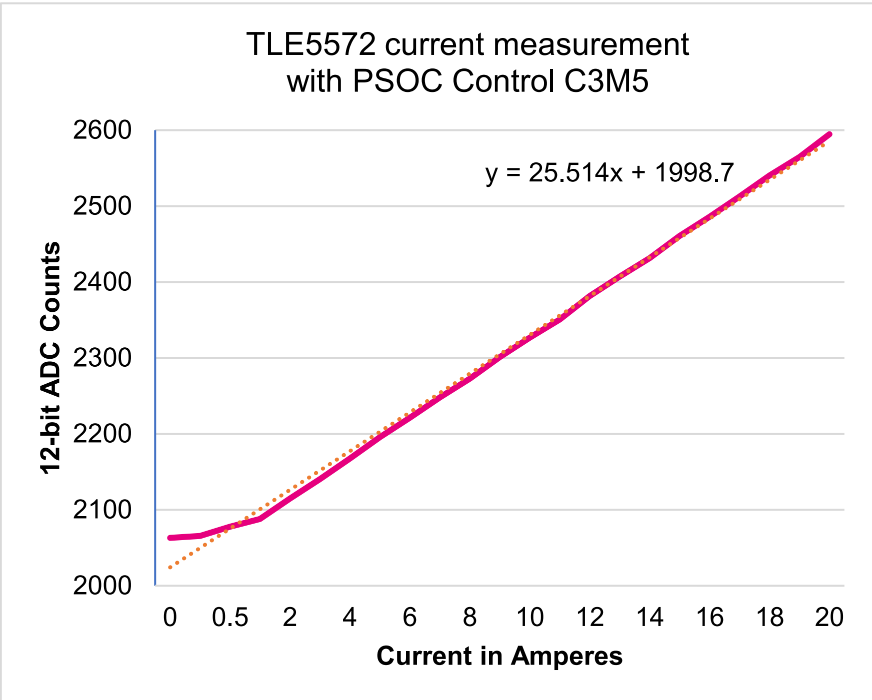


Sensor position on a power rail. Dimensions in millimetres

These two sensors were tested with RAK-GAN and CC1 boards in a laboratory under room conditions at 23°C, 37% humidity. The power rails were gradually loaded to 20A, and the 12-bit ADC values, averaged 8 times, were read from the PSOC Control C3M5 memory for each sensor. The values do not show a linear dependence on current when the current is below 1A. Preliminarily, each sensor has a slightly different sensitivity across devices and boards; therefore, calibration will be required to achieve the required accuracy.



RAK-GaN TLE5571 current vs 12-bit ADC counts



RAK-GaN TLE5572 current vs 12-bit ADC counts

Firmware Overcurrent Detection (OCD) with TLE5571 OCD Output

It is good practice to enable firmware overcurrent detection when working with high-power motors or DC/DC converter loads. The protection level should be set to a lower current limit than the hardware OCP. This ensures you respond quickly to overcurrent events that could damage the components.

In the RAK-GAN board design, the OCD can be read via the P4.3 pin (PWR_OCD), and the OCD threshold is controlled by the MCU-generated PWM signal from the P4.2 pin (PWR_THPWM).

Configuring the right GPIOs in ModusToolbox™ Device Configurator.

The OCD voltage level can be calculated using the equation below:

$$V_{THR} = \frac{OCD_{THR} * S * TF}{10^6}$$

Where:

• V_{THR} is the voltage set externally on the THR pin in [V]. It is recommended to use a non-ratiometric voltage to set the threshold.

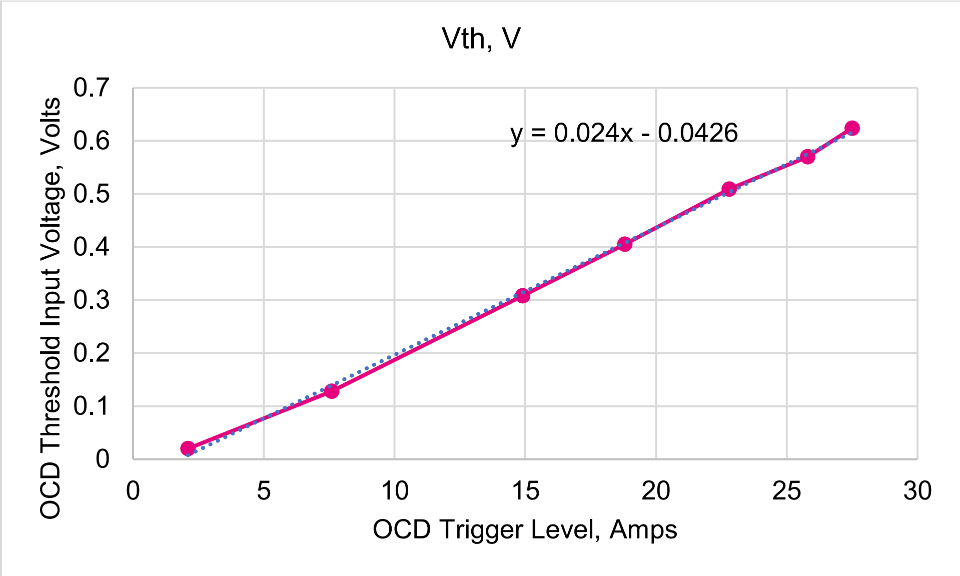
• OCD_{THR} is the desired OCD threshold level in [A]. The threshold shall be between 20% and 180% of the differential magnetic input range full scale.

• S is the sensitivity in [mV/mT].

• TF is the current rail transfer factor in [μT/A].

Since the current rail transfer factor for the TLE5571 was not yet known, the OCD threshold was determined through experimental methods:

OCD Trigger Level, A	Vth, V
2,1	0,02
7,6	0,128
14,9	0,308
18,8	0,405
22,8	0,509
25,8	0,57
27,5	0,624



The TLE5571 OCD trigger voltage level

The results may differ from sensor to sensor; the calibration coefficient is used in the actual [code.](#)[5]:

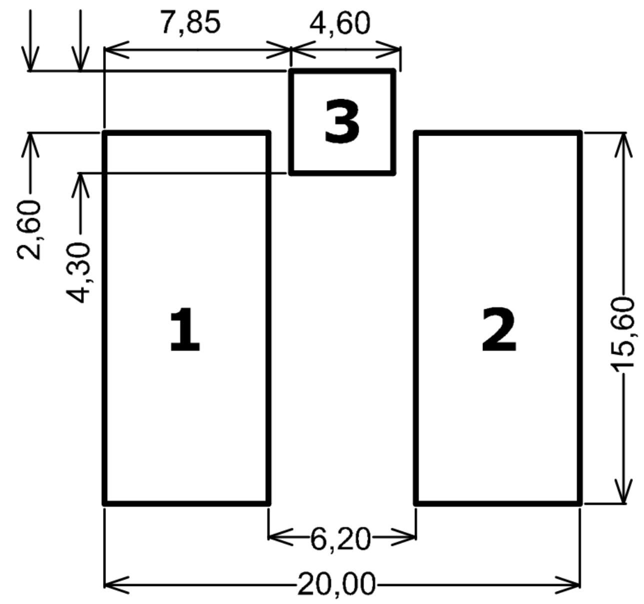
```
#define OCD_TH_RES      (10000)
#define OCD_TH_CAL      (0.0f)
#define OCD_I_AMP       (25.0f) /*Power Input Current Limit*/
```

```
uint32_t OCD_Current_to_PWM(float i_limit)
{
    uint32_t pwr_thpwm = OCD_TH_RES;
    float v_th = 0;

    v_th = 0.024 * i_limit - 0.0426 + OCD_TH_CAL;
    pwr_thpwm = (uint32_t)(v_th / 3.3 * OCD_TH_RES);

    return pwr_thpwm;
} // OCD_Current_to_PWM
```

[5] MCU.c file in the RAK_GAN_DRONE_MOTOR_TEST code example.

Power Inductor Variations

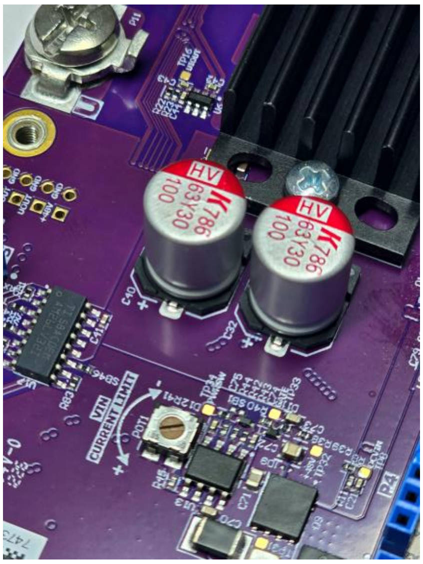
The EPCOS B82559B5472A016 power inductor is mounted on the RAK-GAN board in a factory. But the footprint is designed to accept other types of inductors as well, for example, VISHAY IHLP6767GZER4R7M01 and Pulse Electronics BMMA001717704R7MV1.



Inductor Footprint: Dimensions in millimetres. Note that Pad 3 is not connected

Aluminium Polymer Capacitors for Motor Drive Bridge

Two Yageo KEMET A786MS107M1JLAS015 capacitors are mounted on the RAK-GAN board in a factory (C32, C40), though they are necessary only when 50kHz or lower switching frequencies are used.



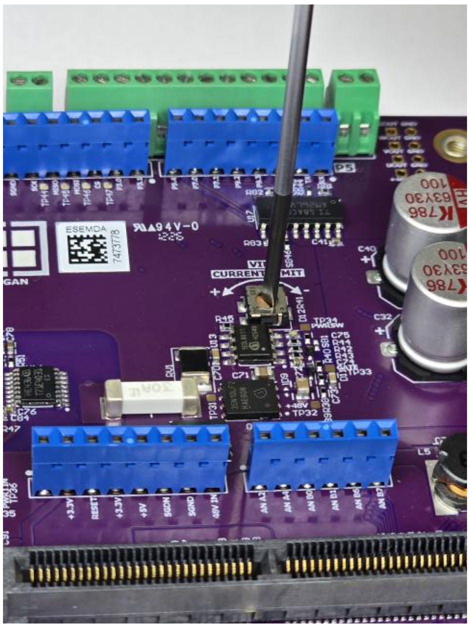
Aluminium Polymer Capacitors C32 and C40

Adjustments of the 7th FET

The ISC035N10NM5LF2 MOSFET Q9, together with the 1EDL8011 MOSFET Driver U13, forms the basis of the 7th FET system that protects the power input from overcurrent and high inrush current events. The current limit and inrush current size depend on the application; they must be adjusted first.

Adjusting the overcurrent limit of the power input

The overcurrent limit is adjusted by turning the potentiometer POT1. Turning clockwise decreases the limit. The current limit can be adjusted to approximately 4A to 30A. The potentiometer POT1 has a nominal value of 1M and is connected in series with the 110K resistor R45, which acts as a maximum current limiter, limiting the adjustable current to approximately 30A. If R45 is replaced with a 0R resistor, or if U13 pin 5 [ILIMIT] is shorted to GND, the overcurrent protection is disabled.



Adjustment of the overcurrent limit for power input

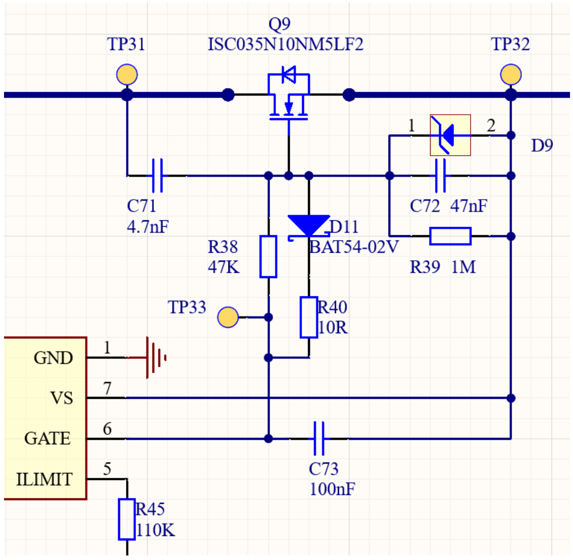
The current limit can be calculated using the equation below. The resistance of the POT1 needs to be measured with a multimeter to ensure the limit is as close as possible to the expected value.

$$I_{LIMIT} = \frac{V_{REF} * R_4}{R_{dson} * k1 * (POT1 + R45)}$$

$$V_{REF} = 1.2V;\ R_4 = k\Omega;\ R_{dson} = 3.5m\Omega; k1 = 1{,}09; R45 = 110k\Omega$$

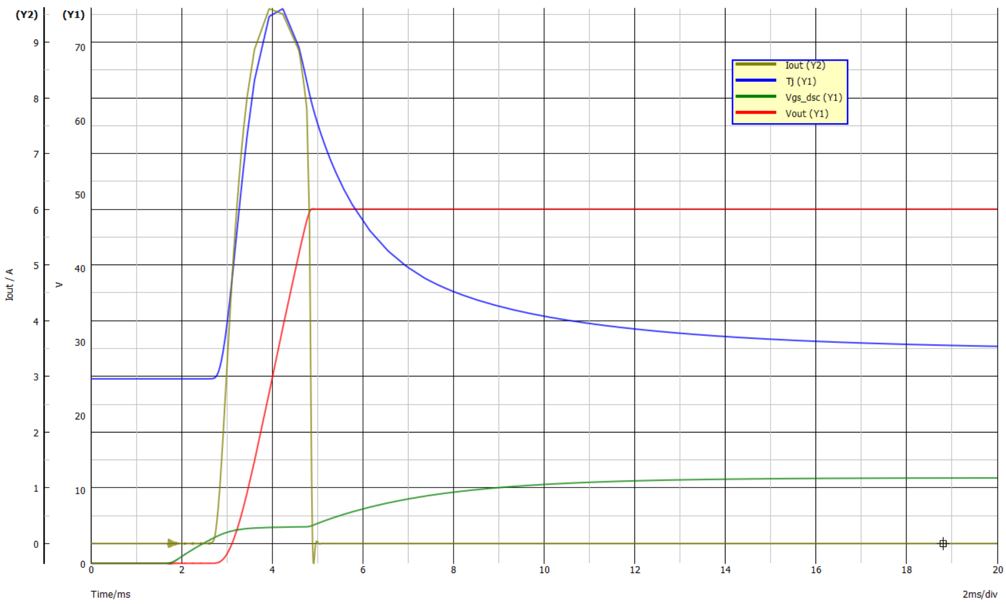
Power input inrush current tuning

The inrush current limiting is tuned by adjusting the C71, C72 & R38 components on the PCB. By default, these are 4.7 nF, 47 nF, and 47 kOhm, respectively, which gives an inrush current of 9.6A with a total load capacitance of 320 µF.



RAK-GaN Power Input inrush current control circuit

The adjustments are made referring to the simulation results. The simulation can be executed using the SIMetrix project located here. In the chart below, the power input inrush current Iout (Y20), the MOSFET junction temperature Tj (Y1), the output voltage Vout (Y1), and the Gate-to-Source MOSFET voltage Vgs_dsc (Y1) are shown.

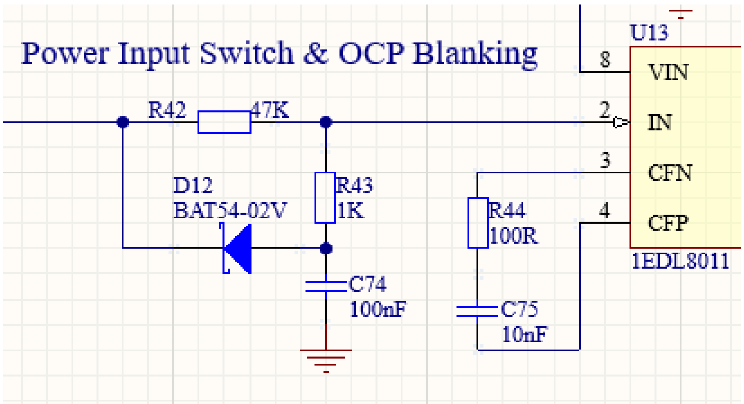


RAK-GaN Power Input inrush current simulation results

The ratio C72/C71 should not exceed 10 times. Larger values will result in higher dV/dt across MOSFET Q9, increasing the risk of induced turn-on (Miller turn-on).

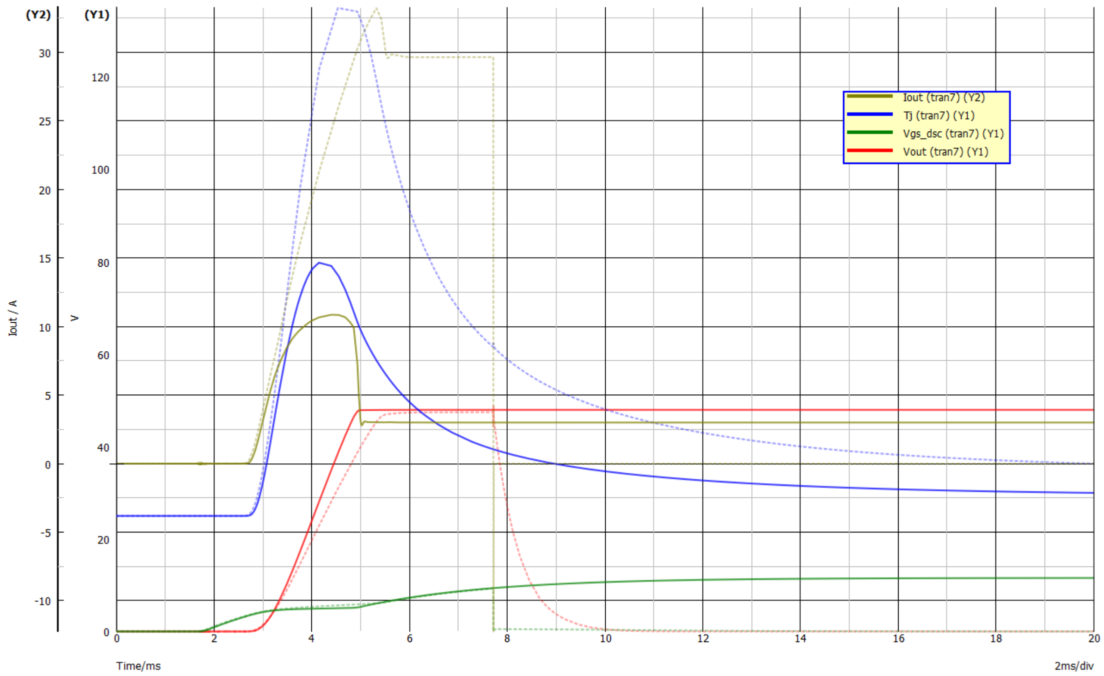
Overcurrent Protection (OCP) blanking for 1EDL8011

The power input startup inrush current must not trigger the overcurrent protection that is handled by the U13 – 1EDL8011 High Side Driver. A method called OCP blanking disables the OCP during system startup. The OCP blanking time is adjusted by varying the values of the external RC network comprising R42, R43, and C74.



RAK-GAN OCP Blanking Circuit

With the given values, the blanking time is approximately 8 ms, as shown in the graph below. The dashed lines show the SIMetrix simulation results with a 1.6 Ohm (30A) load, and the current limit is set to 4A (R_{ILIM} = 1.11MOhm); hence, the OCP triggers after the blanking time. The solid lines represent the results for a 16 Ohm (3A) load and a current limit of 4A (RILIM = 1.11 MOhm); the 10.8A inrush current does not trigger the OCP because it falls within the blanking-time region.



OCP Blanking time simulation (8 milliseconds)

The OCP Blanking time can be calculated referring to Infineon 1EDL8011 datasheet section 5.2.2 „Safe start-up description and OCP blanking“. A total blanking time period consists of two blanking time processes that are calculated using the equations given below:

$$T_{BLANK} = T_{BLANK1} + T_{BLANK2}$$

$$T_{BLANK1=} \; -\tau\left(1+\frac{a*b}{a+b}\right)*\ln\left[\left(1-\frac{V_{LOOP}}{V\_IN}*\left(1+\frac{a}{b}\right)\right)*\left(1+\frac{1}{a}+\frac{1}{b}\right)\right]$$

$$T_{BLANK2} = \frac{V_{REF}}{V_{LOOP}-V_{CBLANK1}}*R*\sqrt{2*C1*Cint*k}$$

$$k = 1.1 + \frac{1\,nF}{1.4*C1}$$

$$V_{CBLANK1} = V_{IN}*\frac{b}{a+b}*\left(1-e^{-\frac{T_{BLANK1}}{\tau*\left(1+\frac{a*b}{a+b}\right)}}\right)$$

$$R = R2;\, a = \frac{R1}{R2};\, b = R_{PD\_IN};\, \tau = R*C1$$

With the RAK-GAN board, the values are:

$Cint := 1\ \mathrm{nF}$	$V\_IN := 3,33\ \mathrm{V}$	$R1 := 47\ \mathrm{k\Omega}$	$R2 := 1\ \mathrm{k\Omega}$	$C1 := 100\ \mathrm{nF}$
$V_{LOOP} := 2,6\ \mathrm{V}$	$R_{PD\_IN} := 11\ \mathrm{M\Omega}$	$V_{REF} := 1,2\ \mathrm{V}$		

And the results are:

$T_{BLANK1ms} = 7,2268\ \mathrm{ms}$	$T_{BLANK2ms} = 1,1674\ \mathrm{ms}$	$T_{BLANKms} = 8,3942\ \mathrm{ms}$

The calculation result closely matches the simulation results. The Smath Solver file for OCP blanking time calculations is available here.

RAK-GaN PCB Design

Designing a typical high-power PCB requires attention to current capacity, safety features, component placement and thermal management of heat-generating components or traces.

Despite the technology of switching transistors used, whether it is BJT, MOSFET or SiC, it is always crucial to deal with certain switching speeds to achieve acceptable system efficiency and comply with EMC laws.

The latest GaN HEMTs are no exception to the previously mentioned aspects. Moreover, these devices have significantly faster switching speeds than MOSFETs; therefore, the PCB design gets more complicated.

Each GaN HEMT application will have its own unique features and specific requirements. Although some common PCB layout design rules can be identified, they are easier to understand when a practical reference design is available. The RAK-GAN development kit from Rutronik System Solutions is selected and analysed as an example project.

Selecting the board's layer stackup

Selecting the right layer stackup is crucial in the very first stages of development. The projects with discrete GaN HEMTs require two outer copper layers to be as narrow as possible to minimise the inductance of the return current path. The smallest possible gap between copper layers is 50 μm, though not all PCB manufacturers may be able to achieve it. Depending on component placement, the gap could be as wide as 150 μm or even 200 μm, but the conductor inductance will increase, degrading performance accordingly.

Because of the requirement for narrow-gapped outer layers, the board will always have at least 4 layers. If a project has a more sophisticated design, the number can be increased until it fulfils the requirements.

The recommended copper thickness is 70 μm (2 oz); if maximum current capacity is not needed, it can be reduced to 35 μm to lower PCB cost.

The RAK-GAN stackup has 6 layers. The Top Layer and the Bottom Layer are power and component layers. Layers 1 and 4 are power-ground (GND) layers for return currents. The two middle layers, Layer 2 and Layer 3, are used for signals and power where possible.

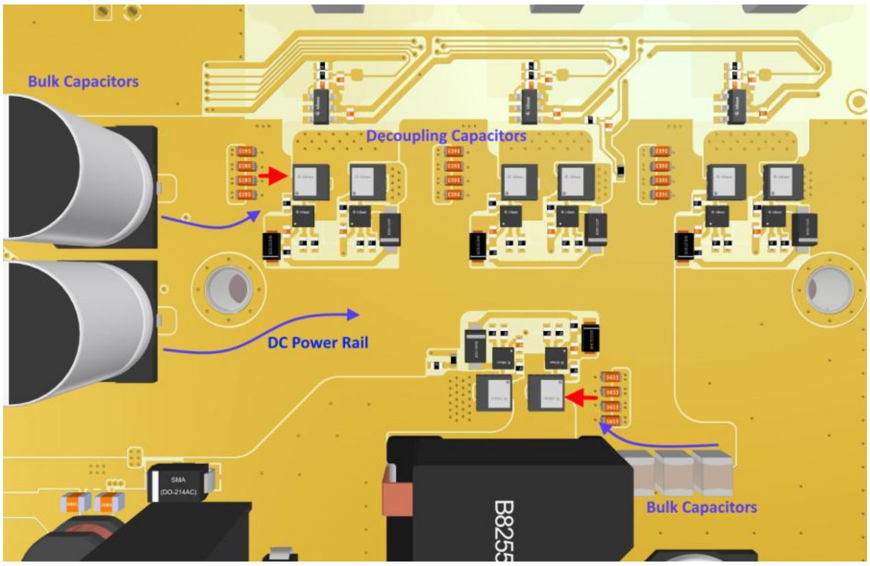
#	Name	Material	Type	Weight	Thickness
	Top Overlay		Overlay		
	Top Solder	Solder Resist	Solder Mask		0.01016mm
1	Top Layer		Signal	2oz	0.07112mm
	Dielectric 2	Core-006	Core		0.1mm
2	Layer 1	CF-004	Signal	2oz	0.061mm
	Dielectric 4	PP-023	Prepreg		0.323mm
3	Layer 2	CF-004	Signal	2oz	0.061mm
	Dielectric 1	Core-034	Core		0.26mm
4	Layer 3	CF-004	Signal	2oz	0.061mm
	Dielectric 5	PP-023	Prepreg		0.323mm
5	Layer 4	CF-004	Signal	2oz	0.061mm
	Dielectric 3	Core-006	Core		0.1mm
6	Bottom Layer		Signal	2oz	0.07112mm
	Bottom Solder	Solder Resist	Solder Mask		0.01016mm
	Bottom Overlay		Overlay		

11 6-Layer PCB stackup

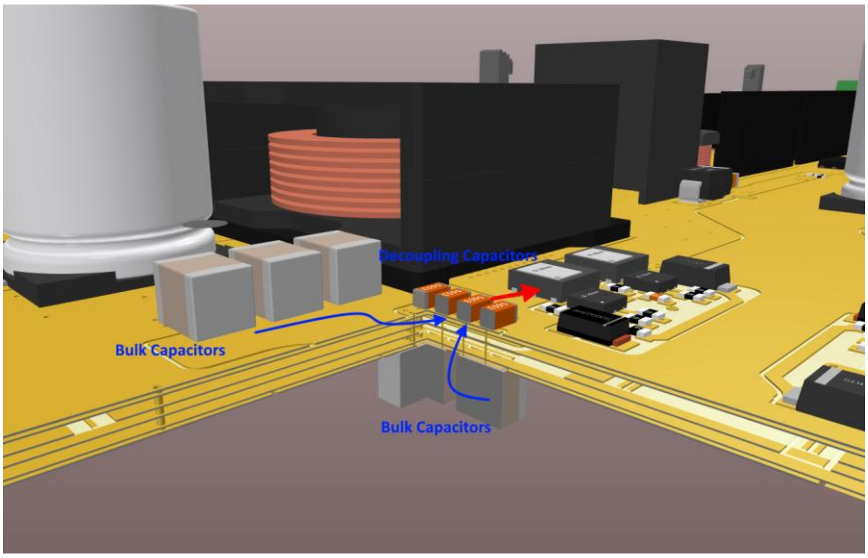
Solving the placement of components

In general, placing components closer together should make the GaN HEMT layout optimal. But this should not be taken at face value. Particular attention must be paid to the positioning of the decoupling capacitors. Their position must be optimal for the ground-return current path, and it must be very close to the GaN bridge's main power input. Besides, the decoupling capacitors are recharged synchronously from bulk capacitors, so they must be placed in close proximity to the decoupling capacitors.

The components on the high and low sides of the bridge drive should be positioned symmetrically to prevent uneven signal delays.



Bulk and decoupling capacitors placement. Top view

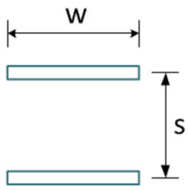


Bulk and decoupling capacitors placement. Section view

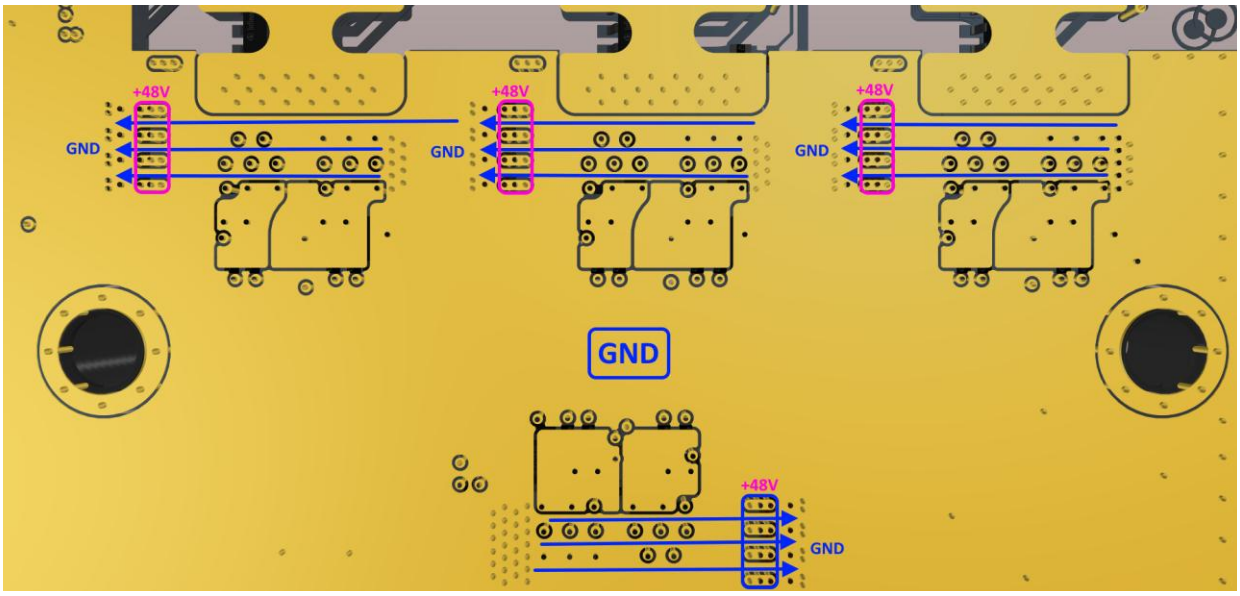
Minimising power and gate loop inductance

Due to very high switching speeds of GaN HEMTs, the inductance of the power loop and gate loop current paths has to be less than 1nH. The proposed method to achieve such inductance is to create two parallel conducting planes with opposite current flow direction. The opposing magnetic fields cancel each other out, reducing electromagnetic interference and lowering inductance. The smaller the gap between the planes, the lower the inductance. The approximate calculation of bi-planar inductivity can be done using the equation below:

$$L = \frac{\mu_0 * s * l}{w}$$



The very first two planes [PCB layers] are critical only for high-speed current flow. The low-speed or constant-current path will flow through the remaining traces on the lower layers of this RAK-GAN PCB. The decoupling capacitors are the primary power source during fast switching in GaN HEMTs; therefore, they need a direct, low-inductance path.



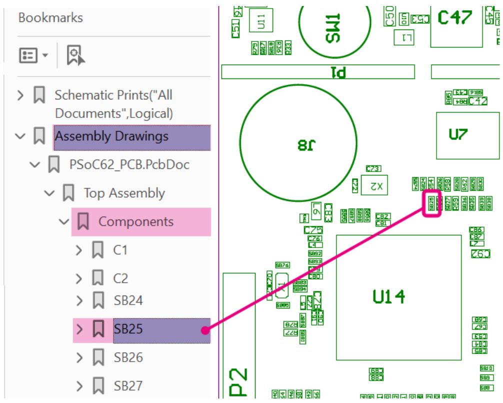
One layer below the top. Ground return current path must be as short as possible

Solder Bridges

Designator	Circuit	Default
SB1	U14 OCD_N to U13 VIN (Overcurrent Shutdown)	Opened
SB2	U14 THR to R50 - R57 voltage divider	Opened
SB3	THPWM signal to U14 THR control	Closed
SB4	DC_HI signal to P4.4, CC1 socket J1	Closed
SB5	DC_LI signal to P4.5, CC1 socket J1	Closed
SB6	U_HI signal to P4.6, CC1 socket J1	Closed
SB7	U_LI signal to P4.7, CC1 socket J1	Closed
SB8	V_HI signal to P9.0, CC1 socket J1	Closed
SB9	V_LI signal to P9.1, CC1 socket J1	Closed
SB10	W_HI signal to P7.2, CC1 socket J1	Closed
SB11	W_LI signal to P7.3, CC1 socket J1	Closed
SB12	PWR_SW signal to P4.0, CC1 socket J1	Closed
SB13	WOC_EN signal to P4.1, CC1 socket J1	Closed
SB14	PWR_THPWM signal to P4.2, CC1 socket J1	Closed
SB15	PWR_OCD signal to P4.3, CC1 socket J1	Closed
SB16	PWR_FLT signal to P7.4, CC1 socket J1	Closed
SB17	USER BUTTON signal to P3.2, CC1 socket J1	Closed
SB18	DC_VOUT signal to AN_A0, CC1 socket J1	Closed
SB19	DC_VIN signal to AN_A1, CC1 socket J1	Closed
SB20		
SB21	PWR_I_IN signal to AN_A5, CC1 socket J1	Closed
SB22	DC_TEMP_P signal to AN_A6, CC1 socket J1	Closed
SB23	DC_IOUT signal to AN_A3, CC1 socket J1	Closed
SB24	W_IOUT signal to AN_B1, CC1 socket J1	Closed
SB25	MD_TEMP signal to AN_B5, CC1 socket J1	Closed
SB26	U_IOUT signal to AN_B2, CC1 socket J1	Closed
SB27	V_OUT signal to AN_B3, CC1 socket J1	Closed
SB28	Ext. Diff. P20-1 signal to AN_B6, CC1 socket J1	Closed
SB29	Ext. Diff. P20-2 signal to AN_B6, CC1 socket J1	Closed
SB30	ARD_IO6 signal to GREEN LED D14	Closed
SB31	ARD_IO2 signal to RED LED D14	Closed
SB32	ARD_IO5 signal to BLUE LED D14	Closed
SB33	U16 STB signal to P3.3	Opened
SB34	U16 CAN TxD to P6.3	Closed
SB35	U16 CAN RxD to P6.2	Closed
SB36	120R CAN Line Termination Resistor	Closed
SB37	U16 STB to GND (Always Enabled)	Closed
SB38	SPI_MOSI signal to SPI_MISO signal	Opened
SB39	HALL_1 signal to P3.3	Closed
SB40	HALL_2 signal to P8.4	Closed
SB41	HALL_3 signal to P8.5	Closed
SB42	DC_TEMP signal to P8	Closed

SB43	POT2 signal to AN_B0, CC1 socket J1	Closed
SB44	U17 ENC_A signal to P7.5	Closed
SB45	U17 ENC_B signal to P7.6	Closed
SB46	U17 ENC_Z signal to P7.7	Closed
SB47	P4 pin 4 Arduino to AN_B1	Opened

The locations of the solder bridges are shown in the 3D model and assembly drawings for RAK-GAN.



How to find a component on the layout

Fuses

The RAK-GAN board has a 30A SMD fuse, F1 (Part No. 0463030.ER) „Littelfuse Inc.“

Changing the Fuses or Solder Bridges

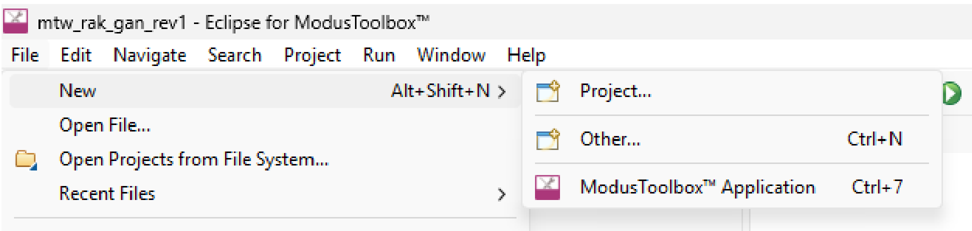
The SMD „Chipping Tool“ is recommended for soldering SMD solder bridges or fuses on the RAK-GAN development board.

Unsoldering the 0402 resistor

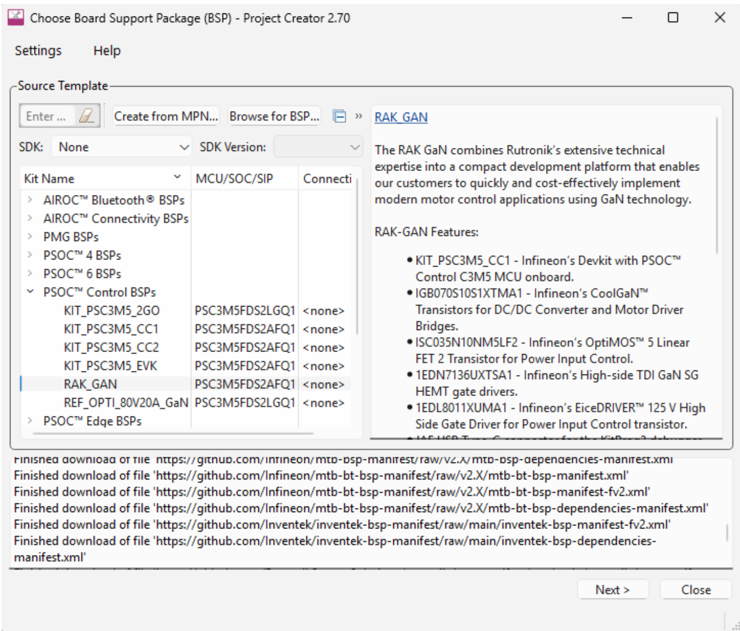
Software and Firmware

Getting Started

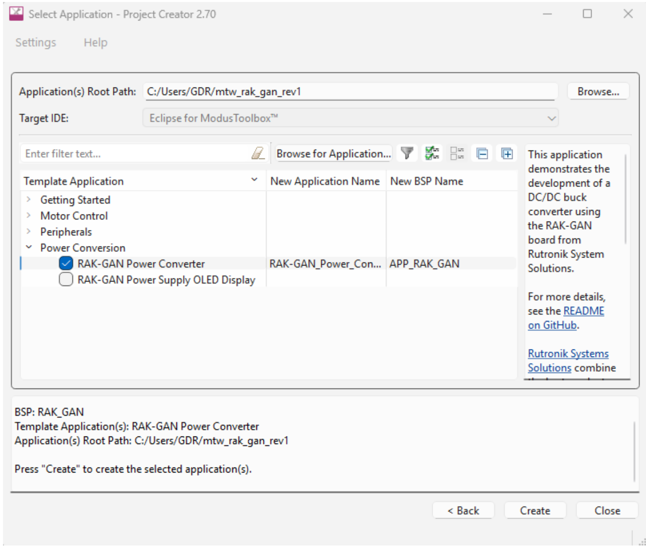
1. Please acquire the RAK-GAN development kit first. Check Rutronik24 or contact [Rutronik System Solutions] by sending a question.
2. Register and/or/and login at the Infineon website (myInfineon tab).
3. Download and install the latest version of ModusToolbox™ software.
4. Plug the KIT_PSC3M5_CC1 board into the J1 socket, which is located on the RAK-GAN board.
5. Connect the KIT_PSC3M5_CC1 USB Type-C socket J1 with your PC; please use your cable.
6. Prepare your power supply, set the output voltage to 10V … 48V and connect the power cables to GND (P18) and VIN (P16). Keep the power supply OFF until the final steps.
7. Run the **Modus Toolbox** application.
8. Go **File** - **New** – **Modus Toolbox Application** and wait for a while.



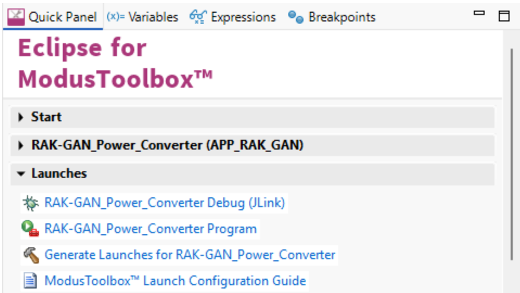
9. Open the **PSOC Control BSPs** block, select **RAK_GAN** and press **Next**.

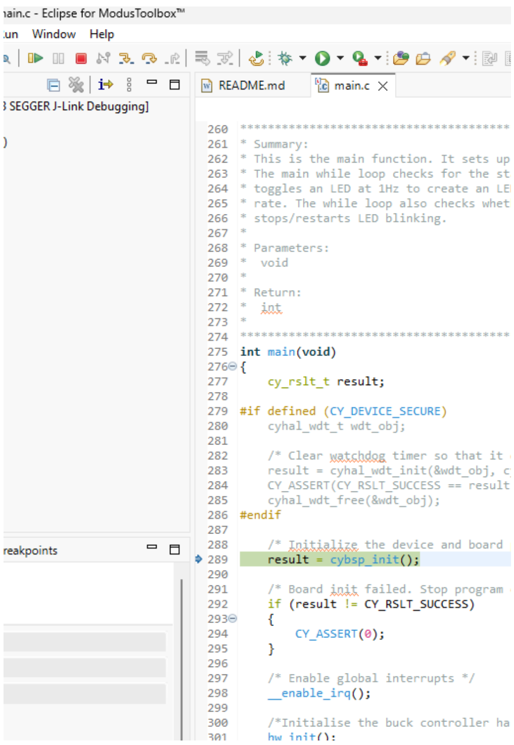


10. Open **the Power Conversion block, check the RAK-GAN Power Converter Application,** and press **Create**.

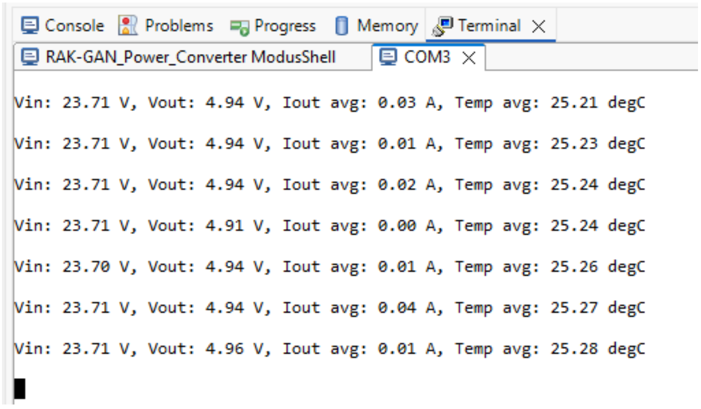


11. **Build** and **Debug** the project. It is recommended to turn the power supply on when the debugging has stopped at “**int** main(**void**)”.





12. The firmware example uses JLink Debug UART for debug output.



Firmware Examples

All these examples can be found on GitHub and are available directly from Project Creator in ModusToolbox™ IDE.

RAK_GAN_DRONE_MOTOR_TEST	This code example was created to test the RAK-GAN with the MIAT8318 100KV motor.
RAK_GAN_POWER_CONVERTER	This application demonstrates the development of a DC/DC buck converter using the RAK-GAN board from Rutronik System Solutions.
RAK_GAN_POWER_SUPPLY_CONTROL	The DC Power Supply application features adjustable voltage and current outputs, which are controlled via a touch-capable OLED screen.
RAK_GAN_I2C_SCANNER	This application is designed to detect all devices connected to the I2C bus.
RAK_GAN_CAN_OBD2	RAK-GAN Code Example for OBDII Protocol.

Starting a new sensorless motor with RAK-GAN and ModusToolbox™ Tools

Please check this document first: ModusToolbox™ Motor Suite Motor Control Library tuning guide

It is recommended to start by using the firmware example that was already previously tested with an actual motor. The current sensor and measurement configuration should already be properly adjusted, so you can focus on configuring and tuning your motor parameters in ParamConfig.h.

```
/*******DC Supply*******/
#define MOTOR_CTRL_VDC_NOM_VOLT                 (48.0f)                    /*[V], Nominal DC bus voltage*/

/*********************************************************************************/
/*******Motor*******/
#define MOTOR_POLE                              (42.0f)                   /*[],  motor poles*/
#define MOTOR_LQ                                (50.0E-6f)                /*[H], Stator q-axis inductance*/
#define MOTOR_LD                                (50.0E-6f)                /*[H], Stator d-axis inductance*/
#define MOTOR_I_AM                              (0.05)                    /*[Wb],  Rotor flux linkage*/
#define MOTOR_R                                 (0.055)                   /*{Ohm],  stator resistance*/
#define MOTOR_TORQUE_MAX                        (6.2f)                    /*[Nm],  maximum torque*/
#define MOTOR_CURRENT_PEAK                      (68)                      /*[A],  peak current rating*/
#define MOTOR_CURRENT_CONT                      (58)                      /*[A],  continuous current rating*/
#define MOTOR_ID_MAX                            (68)                      /*[A], maximum d-axis current*/
#define MOTOR_VOLTAGE                           (48.0f)                   /*[V], motor voltage*/
#define MOTOR_NORM_SPEED                        (4200.0f)                 /*[RPM], nominal speed*/
#define MOTOR_MAX_SPEED                         (5000.0f)                 /*[RPM],  maximum no load speed*/
#if defined(CTRL_METHOD_SFO)
#define MOTOR_MTPV_TORQUE_MARGIN                (90.0f)                   /*[%],  MTPV torque margin*/
#endif
```

Once you have the motor parameters ready, double-check the RAK-GAN board's Hardware and Firmware overcurrent protection limits before connecting and starting the motor using the Motor Suite GUI.

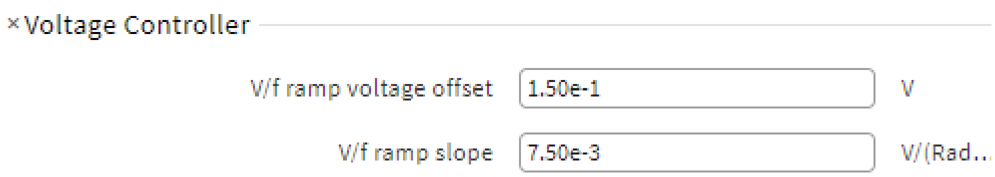
1. The V/F open-loop mode is a good starting point for tuning your sensorless motor in. The Volt_Mode_Open_Loop needs to be selected, and the V/F parameters adjusted for proper motor startup. Normally, this mode is used only to start the motor and accelerate to a speed adequate for Back-EMF generation and closed-loop operation.

Despite the simplicity of V/F open-loop mode, it is not ideal for motor start-up. This mode generates large current ripples that may affect the motor's lifespan. Also, if the mechanical load changes at any point, it needs to be readjusted to compensate.

There are two main parameters to highlight for this mode: the startup voltage offset (v_min) and the voltage-to-frequency ratio (v_to_f_ratio). If any of these parameters is too low, the motor will not start spinning; it will instead shake. If these are too high, the motor will consume too much current and might trigger the overcurrent protection. It may also load the GaN bridge to the point at which the overtemperature protection activates after a certain time. Having this mode in mind is only for starting; it doesn't necessarily mean the parameters are wrong. It depends on how inert your mechanical system is. The adjustment is considered complete when the motor draws a constant phase current across all speed ranges.

One of the fastest ways to adjust the parameters is to program initial V/F mode values into the firmware and then fine-tune them using Motor Suite:

```
/*********Voltage Controller*********/
#define MOTOR_CTRL_VOLT_STARTUP_THRESH          (MOTOR_NORM_SPEED*0.03f)        /*[RPM], startup threshold*/
#define MOTOR_CTRL_VOLT_VF_OFFSET               (0.18f)                         /*[V], V/F ramp voltage offset*/
#define MOTOR_CTRL_VOLT_VF_RATIO                (0.0040f)                       /*[V/(Rad/secz], V/f ramp slope*/
```

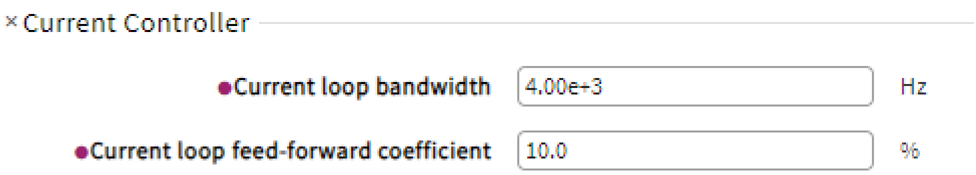


2. The I/F open-loop mode is more complex than the V/F open-loop mode, but it is more beneficial for motor lifespan, and it has a softer transition to closed-loop operation. The most important parameter for this mode to work properly is an open-loop current command (i_cmd_ol). If it is set too low, the motor will shake rather than spin. If it is set too high, it will overload the GaN, and the protections will start triggering.

Also, the current loop controller bandwidth must be adjusted to match the actual performance requirements. A rule of thumb is to have a switching frequency 10–20 times lower. When switching at frequencies of 100kHz and above, this is very difficult to achieve because the microcontroller is not fast enough to respond to changes in every PWM cycle. Please lower the bandwidth or choose a lower PWM-to-fast-loop frequency ratio.

The i_cmd_ol is adjusted in the firmware ParamConfig.c file; it could also be done from the Motor Suite if a variable control tool were added to the GUI Builder for faster adjustment. The current control bandwidth is a common parameter found in ParamConfig.h and in the Motor Suite main GUI.

```
    // Current Control Parameters:
#if defined(CTRL_METHOD_TBC)
    params_ptr->ctrl.curr.bypass = MOTOR_CTRL_CURRENT_BYPASS;
#endif
#if defined(CTRL_METHOD_RFO) || defined(CTRL_METHOD_TBC)
    params_ptr->ctrl.curr.bw = HZ_TO_RADSEC(MOTOR_CTRL_CURRENT_BW); // [Ra/sec], at least one dec below switching frequency
    params_ptr->ctrl.curr.ff_coef = PERC_TO_NORM(MOTOR_CTRL_CURRENT_FF_COEFF); // [#]
    params_ptr->ctrl.curr.i_cmd_thresh = MOTOR_CTRL_CURRENT_STARTUP_THRESH; // [A]
    params_ptr->ctrl.curr.i_cmd_hyst = params_ptr->ctrl.curr.i_cmd_thresh * 0.80f; // [A]
    params_ptr->ctrl.curr.i_cmd_ol = (MOTOR_CURRENT_CONT * 0.20f); // [A]
```



3. Once the V/F, I/F (or any other) motor starting mode is adjusted, the closed-loop tuning can proceed. Beware that closed-loop operation has two loops: a current control loop and a speed control loop, which operate above the current controller. The current control loop must be adjusted with no load on the motor, and the speed loop is adjusted with a loaded motor.

The closed-loop operation engages once the observer activation speed threshold is reached. Usually, it is around 10 per cent of the motor's nominal speed:

```
/*******Observer*******/
#define MOTOR_CTRL_OBS_SPEED_THRESH           (320.0f)      /*[RPM], Observer activation speed threshold*/
#define MOTOR_CTRL_OBS_MIN_LOCK_TIME          (1.0f)                     /*[sec], Observer minimum lock time*/
```

The transition from open-loop to closed-loop operation should be smooth and unnoticeable. If you get a sudden current increase or even an overcurrent event during this transition, you need to adjust the transition coefficient shown below:

```
/*********SPEED Controller*********/
#define MOTOR_CTRL_SPEED_BW                 (40.0f)                  /*[Hz], Speed loop bandwidth*/
#define MOTOR_CTRL_SPEED_OL_CL_TR_COEFF     (45.0f)                /*[%], Open-loop to closed-loop transition coefficient*/
#define MOTOR_CTRL_SPEED_KI_MULTIPLE        (10.0f)                 /*[], Ki multiple for speed looop*/
```

Please be aware that the speed control loop bandwidth affects the motor dynamic response to changes, but it should not exceed 10 times the current control loop bandwidth.

Motor mechanical parameters influence the speed loop control. The Ki and Kp values are automatically calculated from the configuration parameters given below, so please make sure they are adjusted properly before tuning takes place:

```
/*******Mechanical Load*******/
#define MECH_INERTIA                (3.0E-4f)               /*[kg.m^2],  Inertia*/
#define MECH_VISCOUS                (5.5E-4f)               /*[kg.m^2/sec], Viscous Damping*/
#define MECH_FRICTION               (1.5E-1f)              /*[kg.m^2/sec^2], Friction*/

/*******************************************************************************/
```

In practice, these parameters may be difficult to evaluate when a motor is additionally loaded with a mechanical system with high inertia relative to the motor. So, to fine-tune the response, the Motor Suite PID tool is used. The values may be changed at runtime, and the response to the throttle might be observed for the proportional and integral parts of the PI (Kp, Ki). After the values are tuned to optimal, the code might be changed in ParamConfig.c to override automatically calculated Kp and Ki:

```

    // Speed Control Parameters:
#if defined(CTRL_METHOD_RFO)
    if(!params_ptr->autocal_disable.speed_control) /*Skip the calculation if this bit is set*/
    {
      //params_ptr->ctrl.speed.kp = ((8.0f / 3.0f) / (POW_TWO(params_ptr->motor.P) * params_ptr->motor.lam)) * params_ptr->mech.inertia * params_ptr->ctrl.sp
      //params_ptr->ctrl.speed.ki = ((8.0f / 3.0f) / (POW_TWO(params_ptr->motor.P) * params_ptr->motor.lam)) * params_ptr->mech.viscous * params_ptr->ctrl.sp

      /*MIAT 8318 100KV with mounted propeller*/
      params_ptr->ctrl.speed.kp = 0.005079614f;
      params_ptr->ctrl.speed.ki = 0.01179f;

    } // if !speed_control
    params_ptr->ctrl.speed.ff_k_inertia = ((8.0f / 3.0f) / (POW_TWO(params_ptr->motor.P) * params_ptr->motor.lam)) * params_ptr->mech.inertia; // [A/(Ra/sec-
    params_ptr->ctrl.speed.ff_k_viscous = ((8.0f / 3.0f) / (POW_TWO(params_ptr->motor.P) * params_ptr->motor.lam)) * params_ptr->mech.viscous; // [A/(Ra/sec-
    params_ptr->ctrl.speed.ff_k_friction = ((4.0f / 3.0f) / (params_ptr->motor.P * params_ptr->motor.lam)) * params_ptr->mech.friction; // [A]
```

Production Data

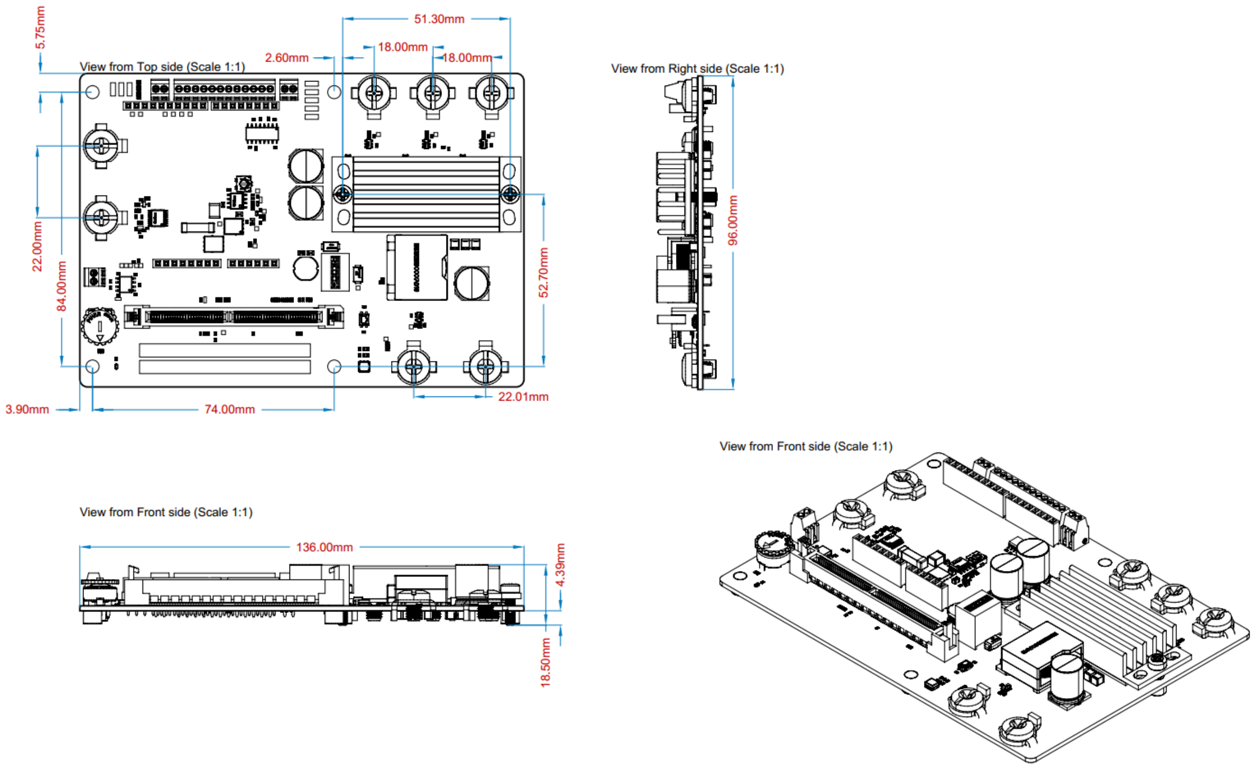
Schematics

You'll find the schematics of RAK-GAN here.

BOM

You'll find the BOM for RAK-GAN here.

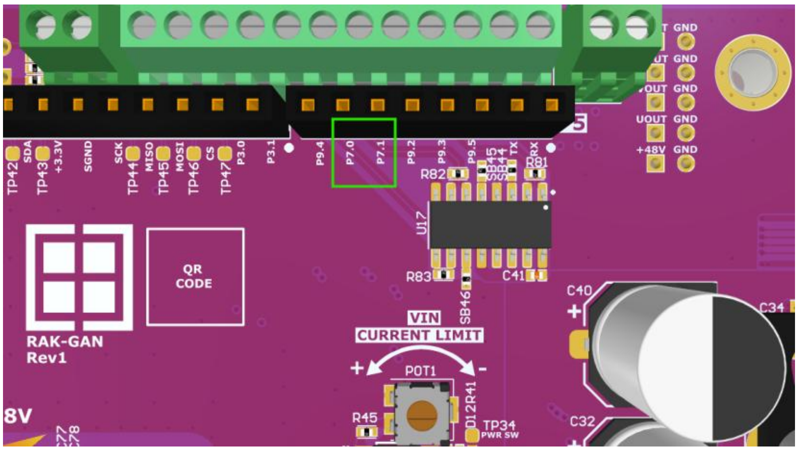
Mechanical Layout



Known Rev1 Issues

Arduino P5 silkscreen label mismatch

The P5 Arduino connector has wrong silkscreen markings. Instead of P6.0 and P6.1, it is marked P7.0 and P7.1.



P22, P23 silkscreen label mismatch

The P22 and P23 “Control Board Pin Output” terminals have two wrong silkscreen markings. Instead of P5.0 and P5.1, it is marked P6.0 and P6.1.